# How HUD Sizes a *221(d)(4) New Construction Mortgage*

A Former Senior MAP Underwriter Explains the Methodology

Generally, the 221(d)(4) is a cost-based program where the loan is sized against what it costs to build, not what the finished project is worth, however, HUD runs three independent tests simultaneously, and the loan is the lowest result across all of them. This article covers how each test works, where BSPRA fits into the math, and why the land value (rather than the land cost) can directly affect loan proceeds. There's a sizing tool at the bottom to run the numbers on your deal.

**Brian Lorenz**
Vice President — Former Senior MAP Underwriter

**Wim Roach**
Vice President

Centennial Mortgage, Inc.
Published April 2026
Last updated May 2026

Current as of May 2026.
Latest version: roachlorenz.com/resources/how-hud-sizes-a-221d4-mortgage/

A note on scope before we start: this article covers **221(d)(4) new construction only**. The 221(d)(4) program also permits substantial rehabilitation of existing properties, but the sizing math is different — construction costs, eligible basis, and the treatment of existing debt all work differently in the sub-rehab context. If you're sizing a substantial rehab, reach out directly and we'll walk through it. Everything below assumes ground-up development.

01

PROGRAM PARAMETERS

## The 221(d)(4) at a Glance

The HUD 221(d)(4) program provides FHA mortgage insurance for the new construction of multifamily properties. It's a single mortgage that covers both the construction period and the permanent loan — at initial closing, the full mortgage amount is insured, construction draws are funded through a HUD-supervised draw process, and the loan converts to the permanent phase at certificate of occupancy. The permanent loan is a 40-year fully amortizing fixed-rate note (meaning no interest rate risk with a perm refinance), non-recourse to the sponsors.

The 221(d)(4) is a cost-based program. The loan is sized against total eligible development cost — not appraised value.

| PARAMETER | HUD 221(D)(4) NEW CONSTRUCTION |
|---|---|
| Loan Basis | **Loan-to-Cost (LTC) of total development cost** |
| Max LTC | **87% (Market Rate) \| 90% (Affordable)** |
| Min DSCR | **1.15x (Market Rate) \| 1.11x (Affordable)** |
| Loan Term | **40 years (permanent phase)** |
| Amortization | **40 years — fully amortizing, no balloon** |
| MIP (Permanent) | **0.25%/yr on outstanding balance** |
| BSPRA | **Available — 10% addition to cost base when eligible** |
| Recourse | **Non-recourse (bad boy carveouts)** |

The 40-year amortization produces a lower annual loan constant than shorter-term programs, which means a given NOI can support a larger loan balance. Combined with the 1.15x minimum DSCR, the 221(d)(4) is structurally more aggressive on proceeds than conventional construction-to-permanent financing, which typically requires 1.25x DSCR and a shorter amortization period (often 25-30 years).

02

CONSTRAINT ONE

## The LTC Test — *Development Cost, Land Basis, and BSPRA*

The LTC test applies 87% (market rate) or 90% (affordable) to the total eligible development cost. Total development cost is not just hardcosts — it's everything required to complete and close the HUD loan: hardcosts, architect and engineering fees, city development fees, third-party report costs, legal fees, title and recording, survey, taxes and insurance during construction, lender and HUD fees, and construction period interest. Note that some "developer costs" are not considered eligible. Things like a development fee (if a for-profit borrower), demolition, and commercial TIs are not eligible and are not included in the mortgageable costs.

One element of the LTC calculation that surprises developers coming from conventional construction lending is the treatment of land. HUD uses the **appraised fair market value** of the land, not the developer's actual acquisition cost. If a developer acquired a site five years ago at $1.5 million and it now appraises at $2.5 million, HUD uses $2.5 million as the land value in the LTC calculation. That increase in land value flows directly into a higher cost basis, which produces a higher LTC-constrained loan amount. The developer effectively captures the appreciation in their land basis as additional loan proceeds.

Builder's and Sponsor's Profit and Risk Allowance (BSPRA) — (read a full, detailed white paper explaining the ins and outs of BSPRA here) — is an addition to total development cost equal to 10% of all project costs excluding land. It's available when the developer has an identity-of-interest with the general contractor (we often make this identity-of-interest artificially if there isn't one already). Mechanically, BSPRA increases cost basis without requiring the developer to spend additional money — because the LTC percentage is applied to that larger cost base, it directly increases the maximum loan amount. On a $28 million development, BSPRA adds roughly $2.5–$3 million to the indicated mortgage because the cost basis used to calculate the LTC loan increased by nearly 10%. If the developer does not have an identity-of-interest with the GC, BSPRA can be replaced by SPRA, which is 10% on everything but land and the hardcosts. The sizing tool below assumes BSPRA eligibility — confirm your deal structure before relying on the LTC figure.

ILLUSTRATIVE EXAMPLE — LTC TEST (WITH BSPRA)

**Hardcosts (inc. Gen. Requirements, Gen. Overhead)** $25,000,000

**Architect Costs** $500,000

**Other Development Fees (Engineering, City Fees, etc.)** $500,000

**Total for All Improvements** **$26,000,000**

**Construction Interest** $1,734,427

**Taxes and Insurance During Construction** $150,000

**3rd Party Reports, Survey, Title & Recording** $135,000

**Legal** $80,000

**HUD and Lender Fees** $742,913

**Total Loan Carrying Costs** **$2,842,340**

**Total Development Cost (Excluding Land)** $28,842,340

**BSPRA 10% Cost Basis Boost** $2,884,234

**Current Land Value** $1,500,000

`Total HUD Eligible Costs = $28,842,340 + $2,884,234 + $1,500,000 =` **$33,226,574**

`87% LTC Loan Amount (rounded down to nearest $100) =` **$28,907,100**

03

CONSTRAINT TWO

## The DSCR Test — *Sizing Against Projected Stabilized Income*

Because there's no operating history on a new construction deal, HUD sizes the DSCR test against the projected stabilized NOI developed by the third-party appraiser — underwritten at current market rents and current operating expenses, not at projected future rents at the time the project leases up. If a developer believes rents will be 15% higher two years from now when the project opens, that expectation doesn't factor into the sizing. HUD sizes against what the market supports today.

The appraiser develops a stabilized income projection with a minimum 7% vacancy for market-rate deals, and replacement reserve deposits are based on the Property Capital Needs Assessment rather than a per-unit estimate. Because the project is new construction, the capital needs assessment will typically produce low replacement reserves — new buildings need less near-term capital expenditure — which means the Debt Service NOI on a new construction deal is often close to the stabilized market NOI. HUD's vacancy floor applies regardless of how tight the market is or how strong lease-up projections look.

To size the DSCR-constrained loan: multiply the Debt Service NOI by 0.87 (the arithmetic equivalent of 1.15x minimum DSCR) to get the maximum allowable annual debt service, then divide by the loan constant. The loan constant is the sum of the interest rate, the annual permanent MIP (0.25%), and the amortization rate at the note rate over 40 years. The 40-year amortization produces a lower constant than shorter-term programs, which means a given NOI budget supports a larger loan balance.

ILLUSTRATIVE EXAMPLE — DSCR TEST

**Debt Service NOI:** $2,500,000 | **Rate:** 6.00% | **Amortization:** 40 years | **Permanent MIP:** 0.25%/yr

Max Annual Debt Service = $2,500,000 × 0.87 = **$2,175,000**

Amortization rate (at 6.00%, 40yr) = annual P&I per $1 − 6.00% = **0.6026%**

Loan Constant = 6.00% (interest) + 0.25% (MIP) + 0.6026% (amortization) = **6.8526%**

DSCR Loan Amount = $2,175,000 / 0.068526 = **$31,739,946**

At these assumptions, the DSCR test produces more proceeds than the LTC test from the prior example ($28.9M). The LTC test is the binding constraint here — the DSCR ceiling is not the limiting factor. If NOI were lower — say $1,800,000 — the DSCR-constrained loan would be roughly $22.9 million, which would fall below the LTC ceiling and make DSCR the binding constraint instead.

Higher rates produce a higher loan constant, which reduces the DSCR-constrained loan amount for the same NOI. There's an additional rate channel unique to construction loans: construction period interest is a variable cost that scales with both the loan amount and the interest rate. As rates rise, construction interest as a percentage of the loan increases, which also pushes up the variable cost component of the LTC calculation. Both tests tighten simultaneously as rates move higher.

04

CONSTRAINT THREE

## The HUD Statutory Mortgage Limit — *Per-Unit Dollar Caps*

HUD publishes per-unit dollar limits by bedroom count that establish an absolute ceiling on the loan, independent of both development cost and income. The base statutory limits for new construction are set by statute and are generally higher than the limits that apply to refinance programs, reflecting the higher per-unit cost of delivering new construction relative to the stabilized value of existing properties.

The High Cost Percentage (HCP) factor is applied uniformly at 270% across all markets, multiplying the base per-unit limits upward. To the statutory unit calculation, HUD adds two additional items: Cost Not Attributable to residential use (estimated at 15% of hardcosts for new construction), and 100% of the appraised land value. Cost Not Attributable covers commercial space, community facilities, and non-residential building components.

ILLUSTRATIVE EXAMPLE — STATUTORY TEST (ELEVATOR BUILDING)

178-unit elevator property with 66 one-bedroom, 23 two-bedroom, and 89 three-bedroom units:

1-BR: 66 units × $84,706 × 2.70 = **$15,094,609.20**

2-BR: 23 units × $103,004 × 2.70 = **$6,396,548.40**

```
3-BR: 89 units × $133,252 × 2.70 = $32,020,455.60

Unit Total = $53,511,613.20

Cost Not Attributable (15% × $25,000,000 hardcosts) = $3,750,000.00

Land Value = $1,500,000.00

Statutory Limit = $53,511,613.20 + $3,750,000 + $1,500,000 = $58,761,613.20
```

On most new construction deals, the statutory limit is not the binding constraint. The per-unit limits — particularly at 270% HCP — are generous enough that LTC or DSCR will produce the lower loan amount in the vast majority of markets. The exception is very large-unit properties in high-rent metros where income and development costs would otherwise support a very large loan. Worth running the calculation, but it should not drive early-stage deal decisions in typical markets.

05

PUTTING IT TOGETHER

## Which Test Bites — *and When*

The actual loan amount is the lowest result across all three tests. In practice:

→ **Normal construction costs, low market interest rates —**

LTC is typically the binding constraint. DSCR on a new construction deal — sizing against current stabilized rents at a low rate — frequently supports a larger loan than the LTC ceiling allows. The developer's cost structure is what limits the loan.

→ **Normal construction costs, high market interest rates —**

DSCR is more likely to bind. Higher rates push up the loan constant, reducing the income-supported loan amount while the construction budget has not necessarily decreased correspondingly.

The tool below runs all three mortgage amount tests simultaneously so you can see which one is binding and by how much. Changing the construction costs, adjusting the NOI, or adjusting unit counts will show you in real time how the binding constraint shifts.

06

FREQUENTLY ASKED

# Common Questions About *221(d)(4) Sizing*

**How does 221(d)(4) loan sizing differ from 223(f) sizing?**

The 221(d)(4) is cost-based; the 223(f) is value-based. A 223(f) runs an LTV test against an appraisal of an existing property, plus a cost-of-refinance criterion. A 221(d)(4) has neither — it runs an LTC test against total development cost, and there is no refinance to cost out. The 221(d)(4) also amortizes over 40 years rather than 35, which produces a lower loan constant and lets the same NOI support a larger loan. And because there's no operating history, the DSCR test runs on the appraiser's projected stabilized NOI at current rents and expenses rather than on actual results.

**What is BSPRA and how does it affect 221(d)(4) loan sizing?**

BSPRA — the Builder's and Sponsor's Profit and Risk Allowance — is a 10% addition to the development cost base excluding land, available when the developer is related to the general contractor. Because the LTC percentage is applied to total development cost, adding BSPRA raises the base that 87% is calculated against, which raises the LTC-constrained loan amount.

**How does construction period interest factor into the sizing?**

Construction interest is a variable cost inside the LTC calculation, estimated as roughly half the loan balance multiplied by the interest rate over the construction term. The half-balance convention reflects the draw structure — draws fund progressively, so the average outstanding balance across the term is about 50% of the final loan amount. Because that cost scales with both the loan amount and the rate, it creates a second rate-sensitive term in the LTC test: higher rates increase the construction interest escrow, which increases total development cost, which increases the LTC-constrained mortgage amount.

**Does HUD require a third-party appraisal on a new construction deal if there's no appraised value test?**

Yes. HUD commissions an independent appraisal on every 221(d)(4). The appraiser establishes the project's NOI, which feeds the DSCR test, and provides the appraised land value used in the LTC and statutory calculations. The appraisal doesn't set a ceiling on proceeds the way an LTV test would, but it's a required third-party report and its outputs run through the sizing.

**Is the 221(d)(4) available for affordable housing with tax credits?**

Yes, and it's commonly used in LIHTC transactions. Affordable deals qualify for the 90% LTC ceiling and the 1.11x minimum DSCR rather than 87% and 1.15x. The interaction between the 221(d)(4) and a tax credit equity structure brings its own HUD requirements around equity pay-in timing and credit enhancer approvals, which are beyond the scope of this article — reach out if you're working on a tax credit deal.

---


## Wim Roach & Brian Lorenz

HUD/FHA Multifamily Origination — Centennial Mortgage, Inc.



Centennial Mortgage, Inc. is an approved HUD/FHA MAP lender. All loan sizing results produced by this tool are hypothetical estimates based on user-supplied assumptions and are not a commitment to lend, a loan application, or underwriting analysis. HUD program parameters, statutory limits, and MIP rates are subject to change. Contact Wim Roach & Brian Lorenz directly for current rate guidance and deal-specific analysis. Wim Roach & Brian Lorenz are loan originators with Centennial Mortgage, Inc.